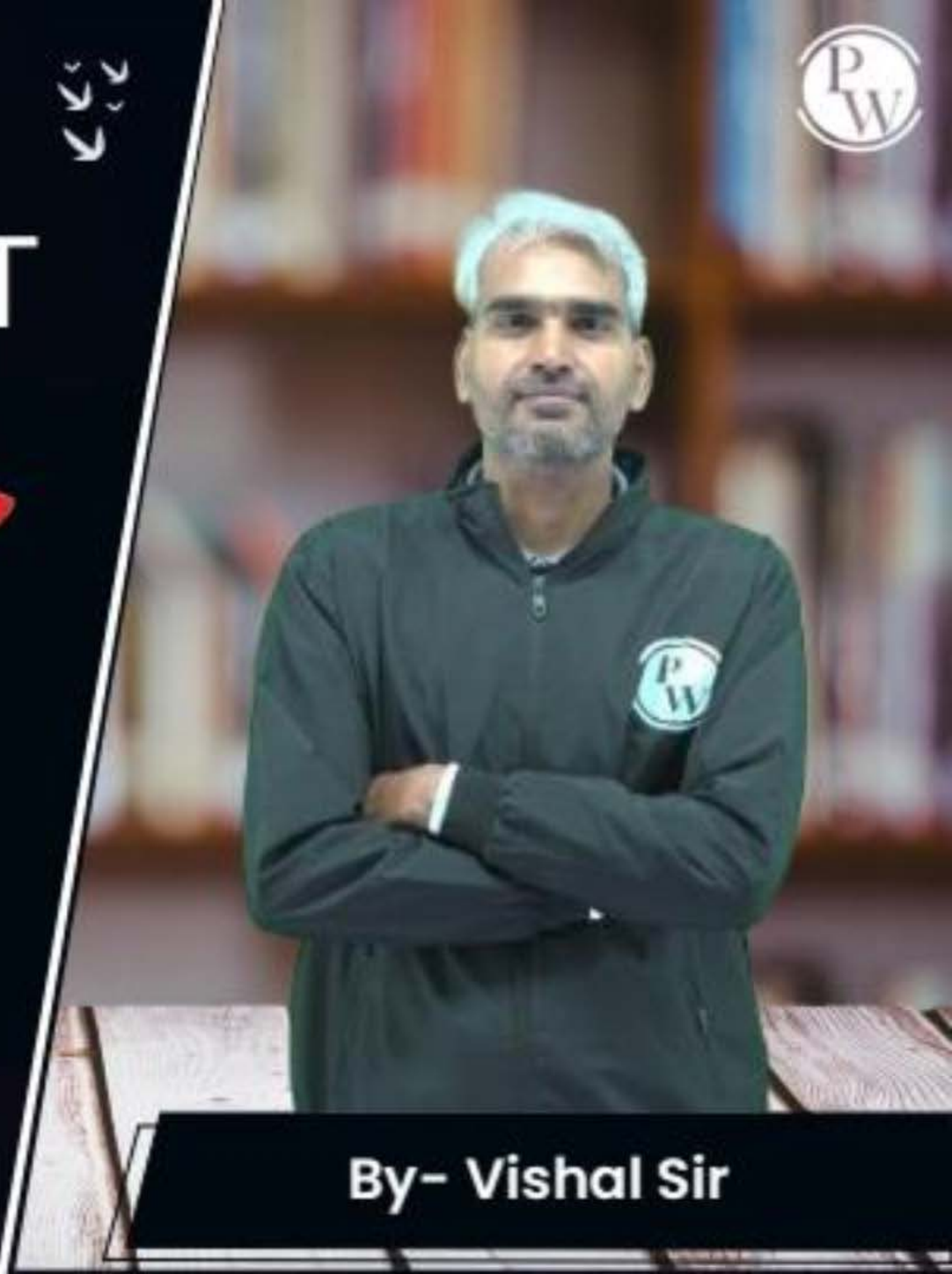
Computer Science & IT



Compiler Design

Syntax Analysis(Parser)

Lecture No. 14



By- Vishal Sir

Recap of Previous Lecture



Topic — Construction of DFA of LR(1) items

Topic — CLR(1) Parser

Topic — CLR(1) Grammar

Slide

Topics to be Covered

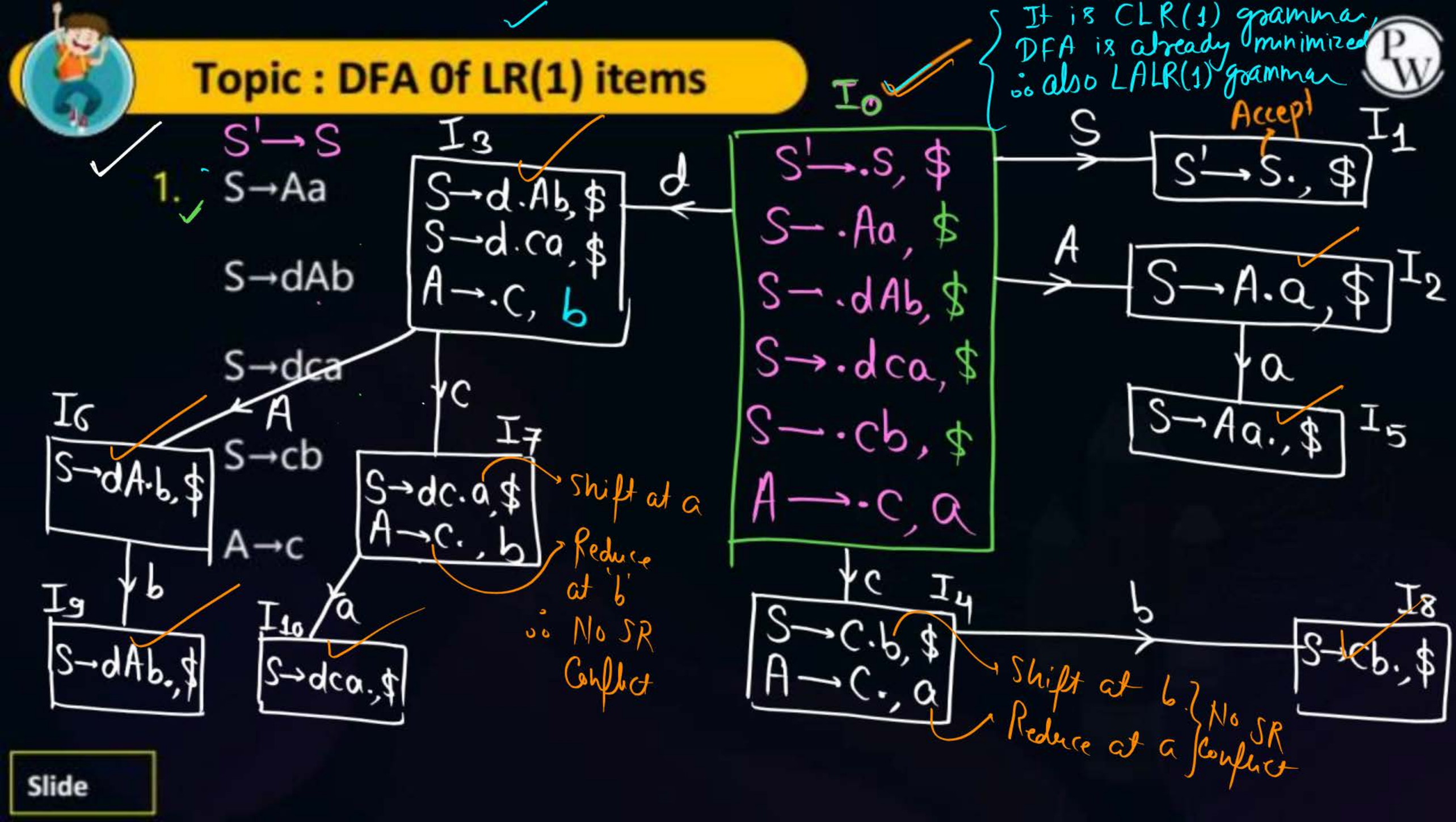


Topic LALR(1) Parser

Topic LALR(1) Grammar

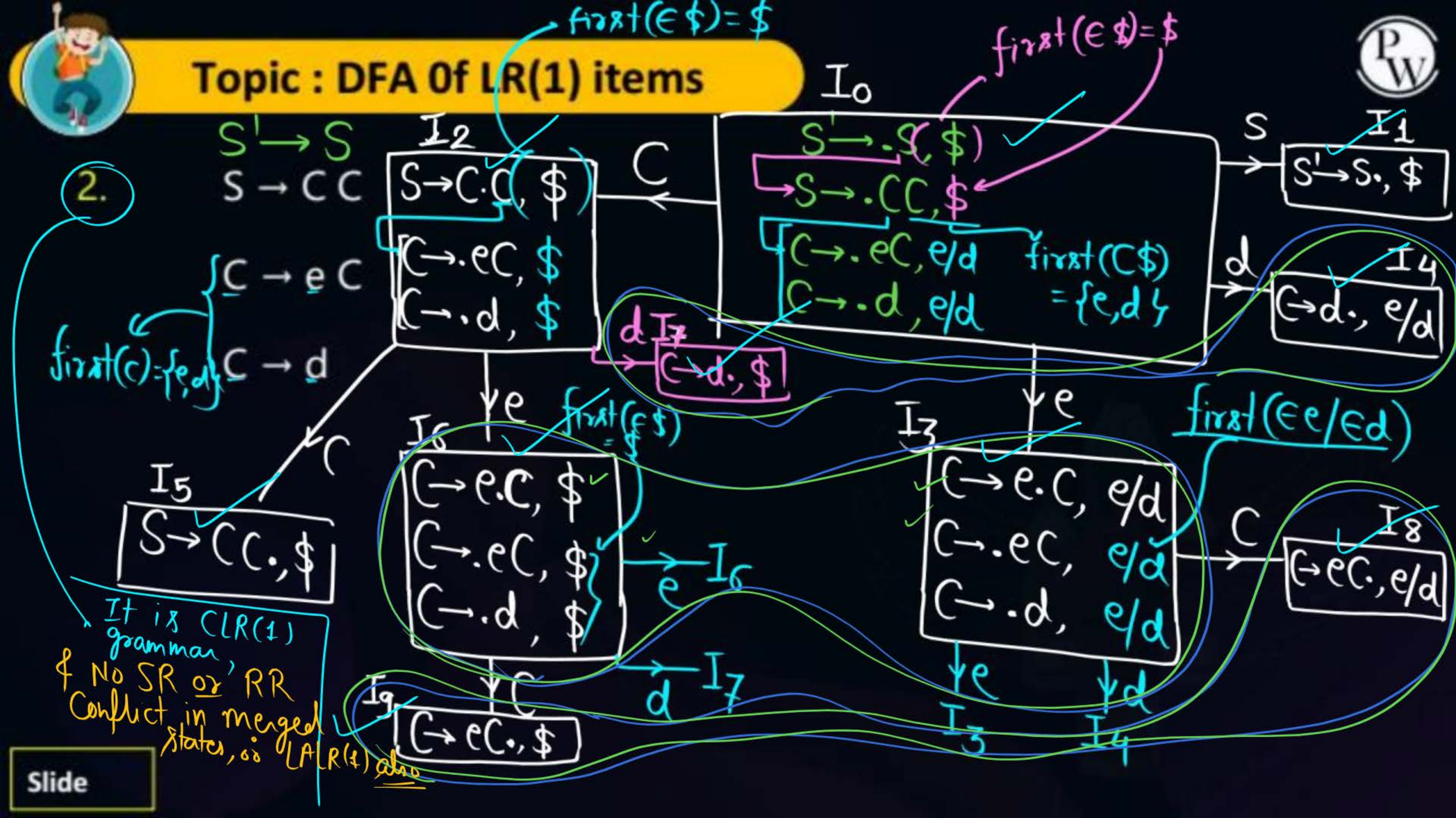
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Topic : DFA Of LR(1) items

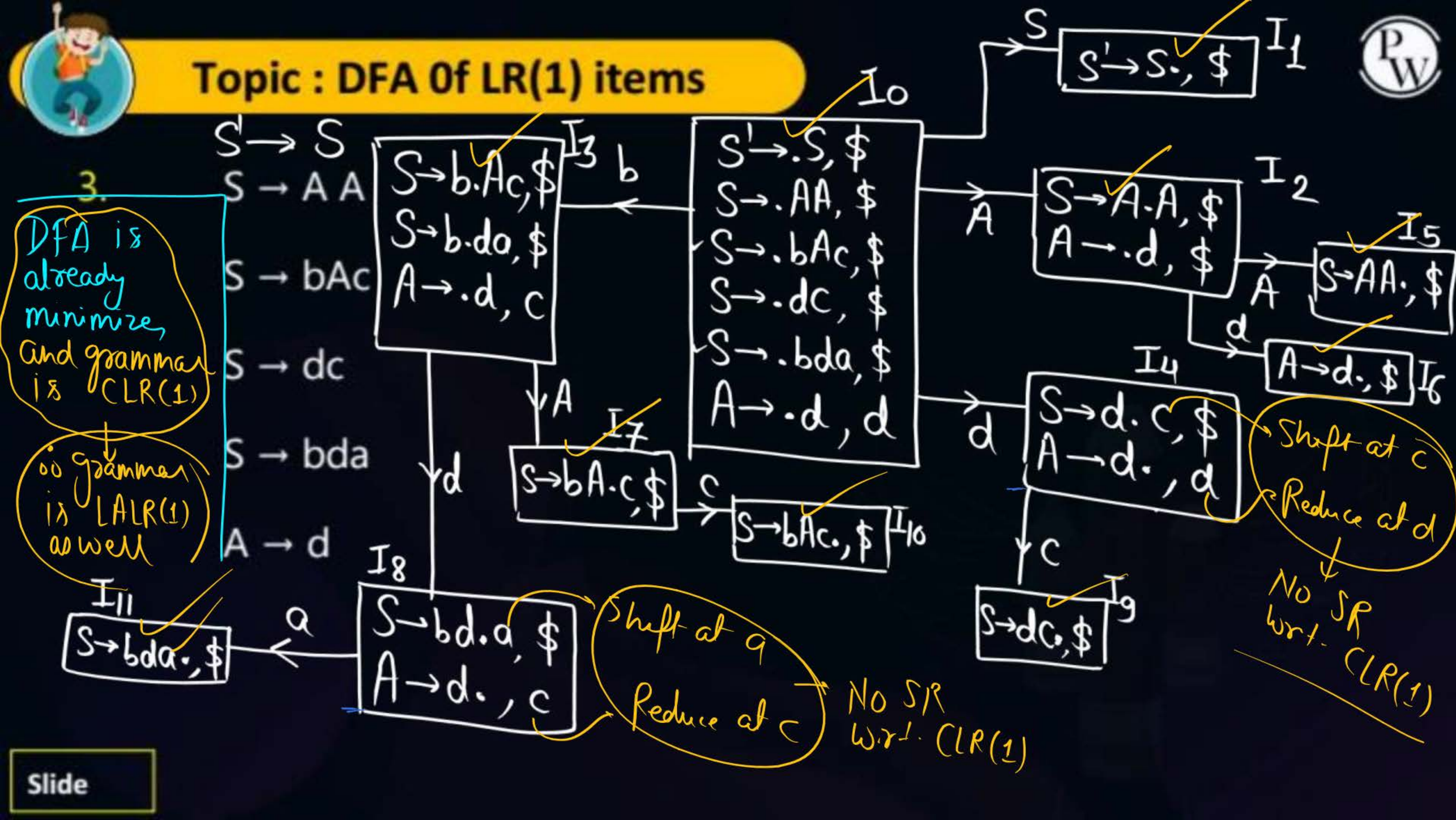


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Topic : DFA Of LR(1) items

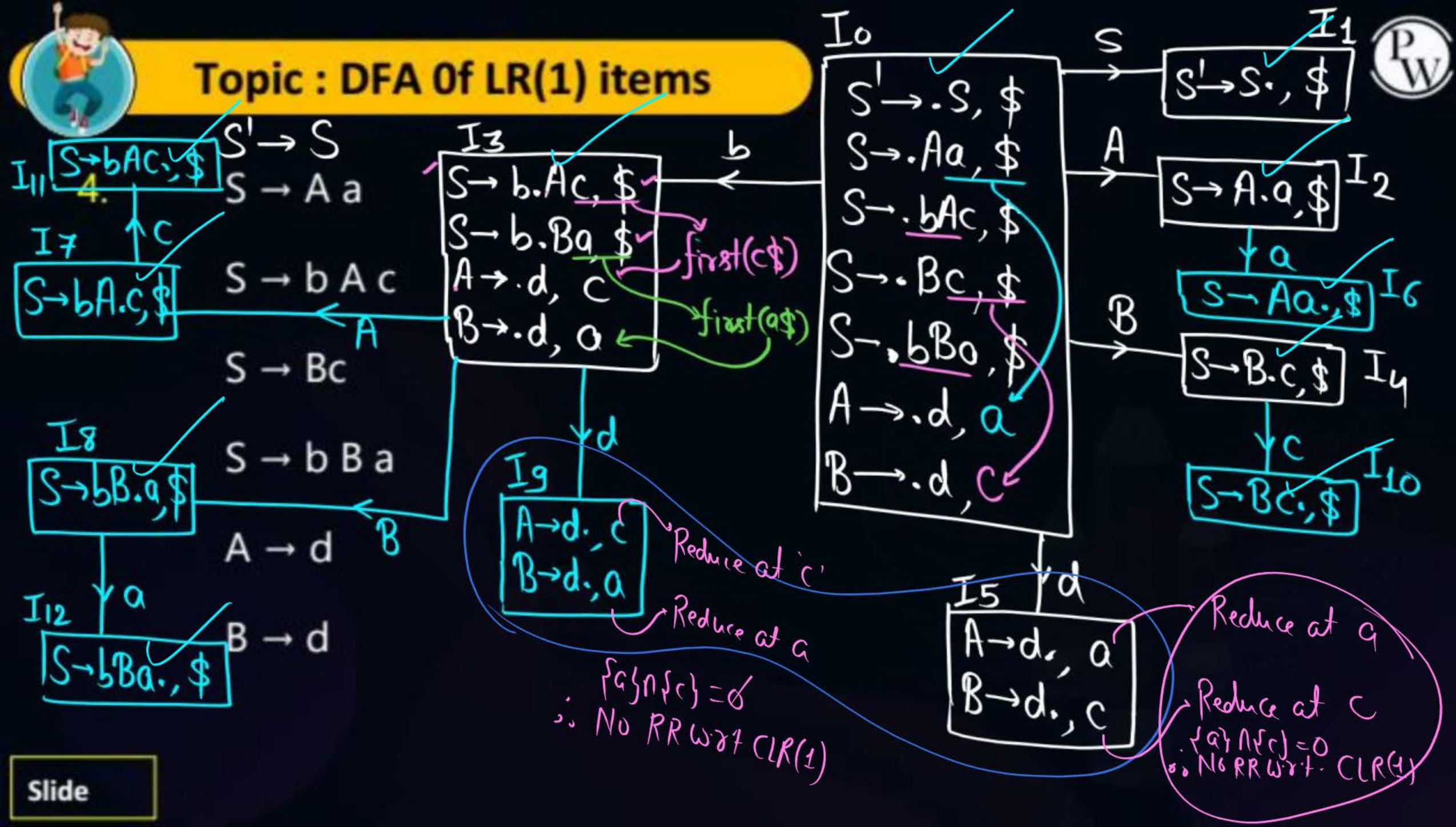


Topic : DFA Of LR(1) items



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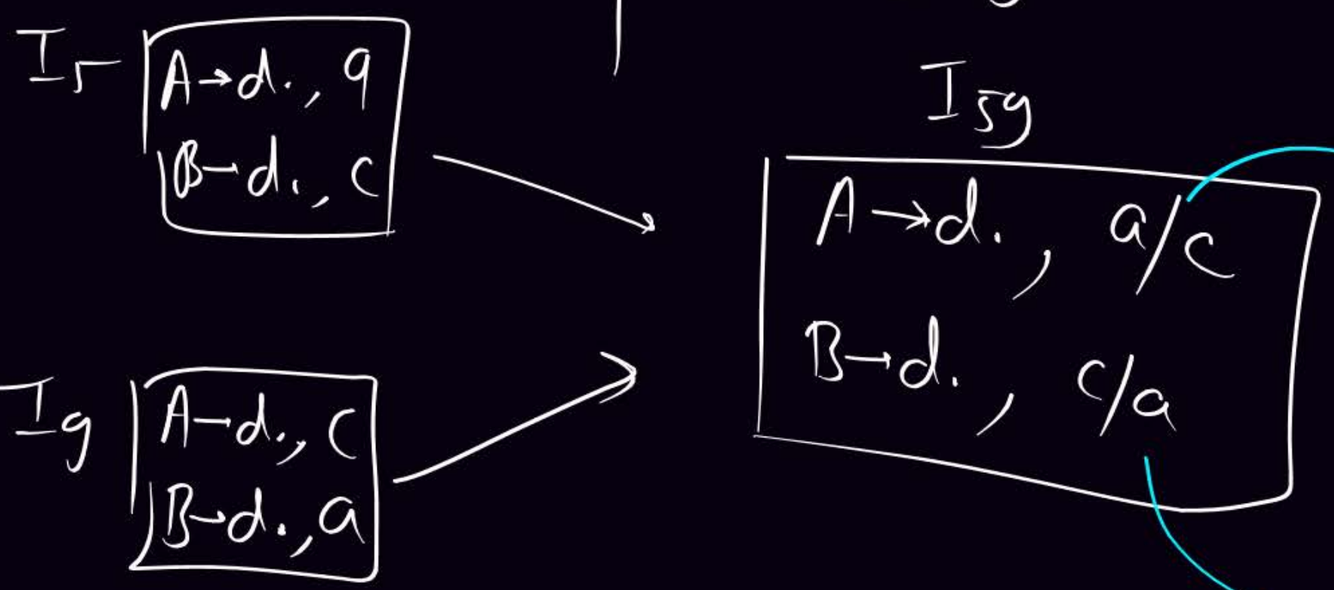
Topic : DFA Of LR(1) items



Slide

In above DFA I_5 & I_9 can be merged

after merging



$\{a,c\} \cap \{c,a\} = \{a,c\} \neq \emptyset$

∴ RR Conflict w.r.t. LALR(1)

∴ Grammar is CLR(1) but not LALR(1)



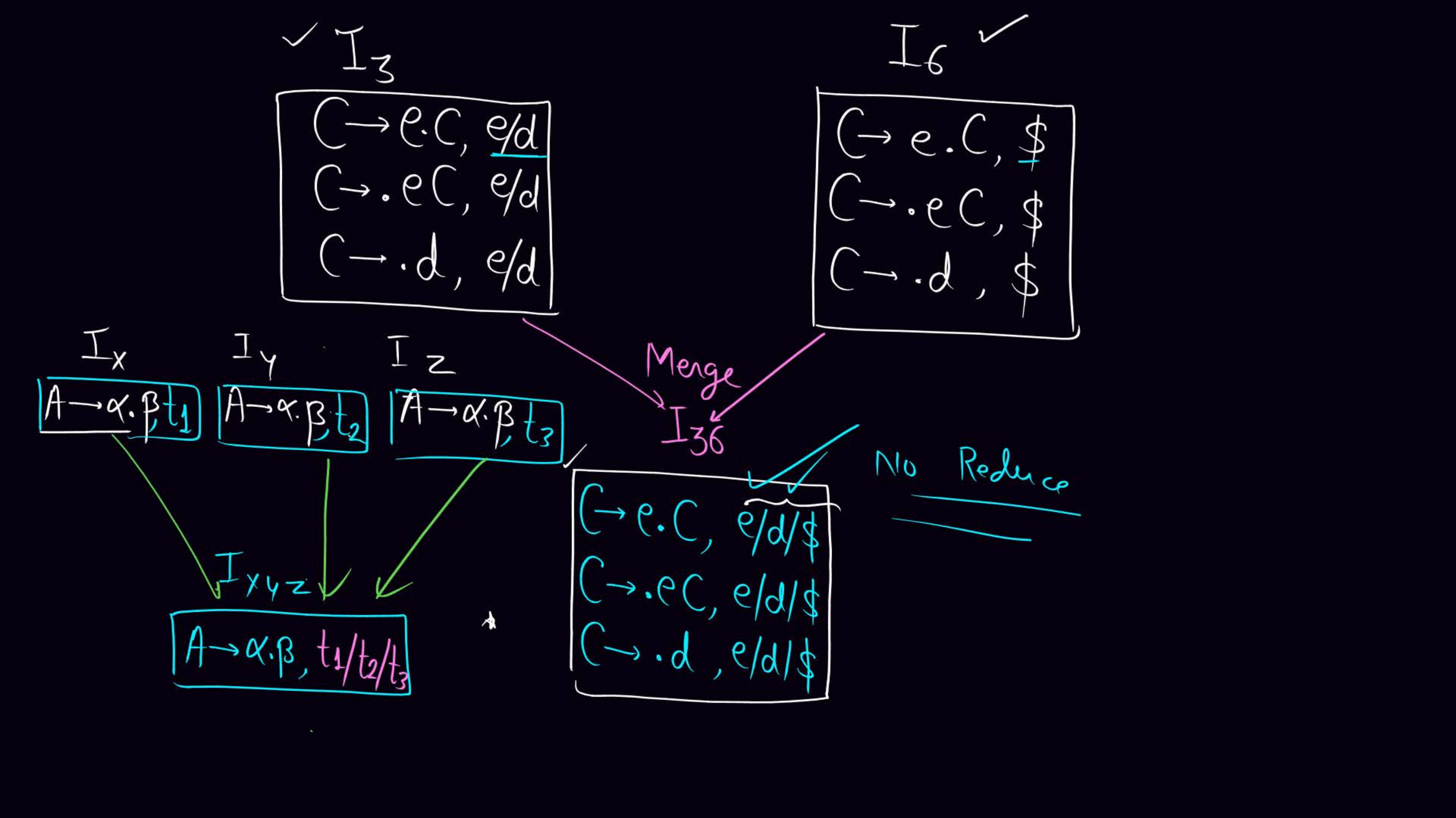
Topic : LALR(1) Parser

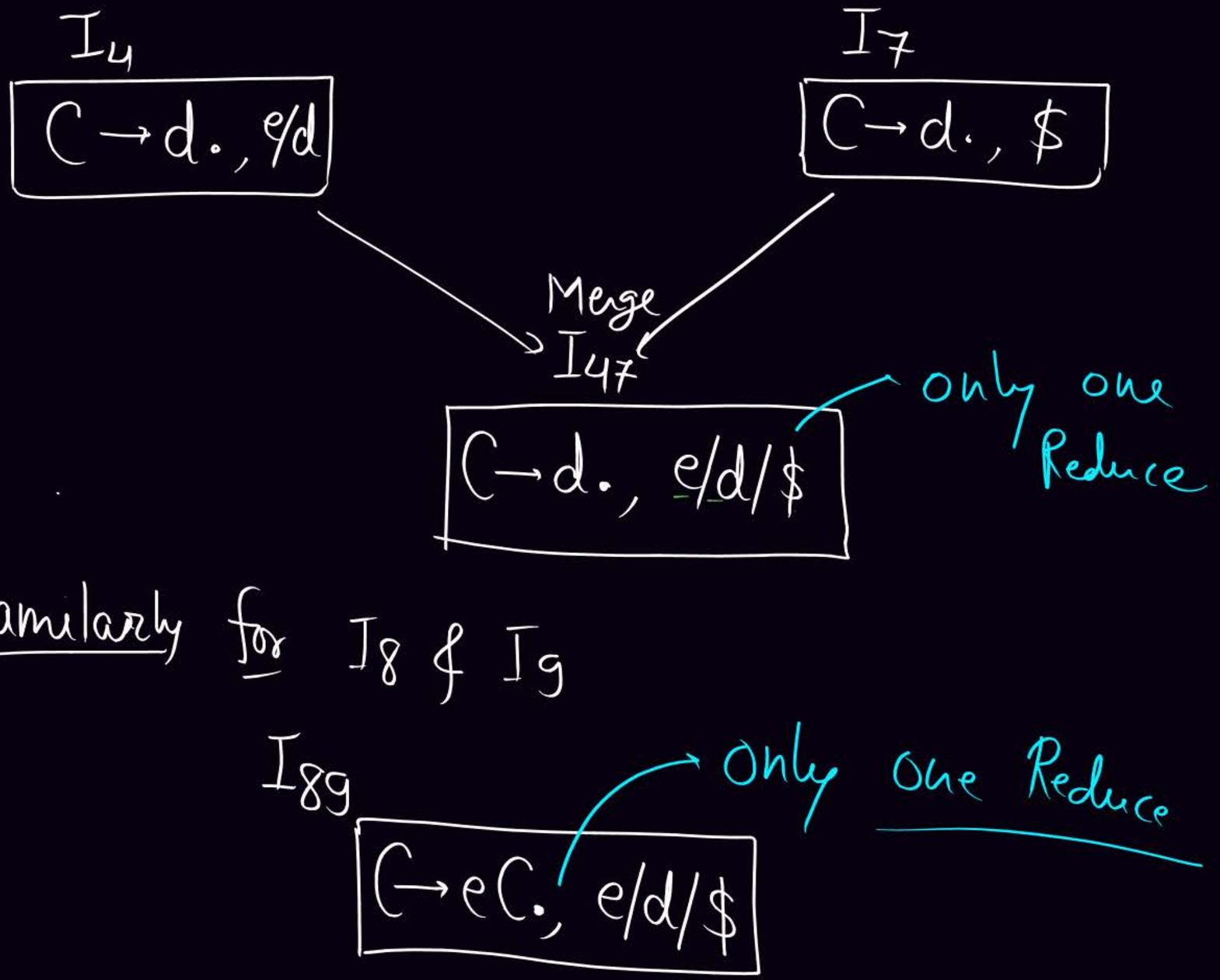
* The parser table Constructed for CLR(1) parser may contain more number of states, ∴ Size of CLR(1) parser table may be too large.
* LALR(1) parser will try to reduce the size of parser table, while not loosing too much in terms of flexibility

Slide

* For LALR(1) parser, the states of the dfa of LR(1) items with same Core-set are merged into a single state.

* While merging the states the look-aheads will be unioned


I3
C → e.C, e/d
C → .eC, e/d
C → .d, e/d
I6
C → e.C, $
C → .eC, $
C → .d, $
Merge
I36
C → e.C, e/d/$
C → .eC, e/d/$
C → .d, e/d/$
No Reduce
Ix
A → α.β, t1
Iy
A → α.β, t2
Iz
A → α.β, t3
Ixyz
A → α.β, t1/t2/t3


I4
C → d., e/d
I7
C → d., $
Merge
I47
C → d., e/d/$
only one Reduce.
Samilarly for I8 & I9
I89
C → eC., e/d/$
only one Reduce



Topic : LALR(1) Parser

1. Obtain the augmented grammar.
2. Construct the DFA of LR(1) items.
3. Replace the states of the DFA of LR(1) items having the same core set with their union. {i.e., Merge the states with same core-set}
4. After merging the states construct LALR(1) parsing table using the following rules.



Topic : ACTION Part w.r.t. LALR(1) Parser

After Merging the states,

1. Shift Action : Same as CLR(1)

2. Reduce Action : If state "i" of the dfa includes an item of the form "A→α . , t1/t2...", where A→α is the jth production of grammar, then update

   ACTION [i , n] = Rj For all n ∈{t1,t2,...}

3. Accept :
4. Error :

} Same as CLR(1)

Slide



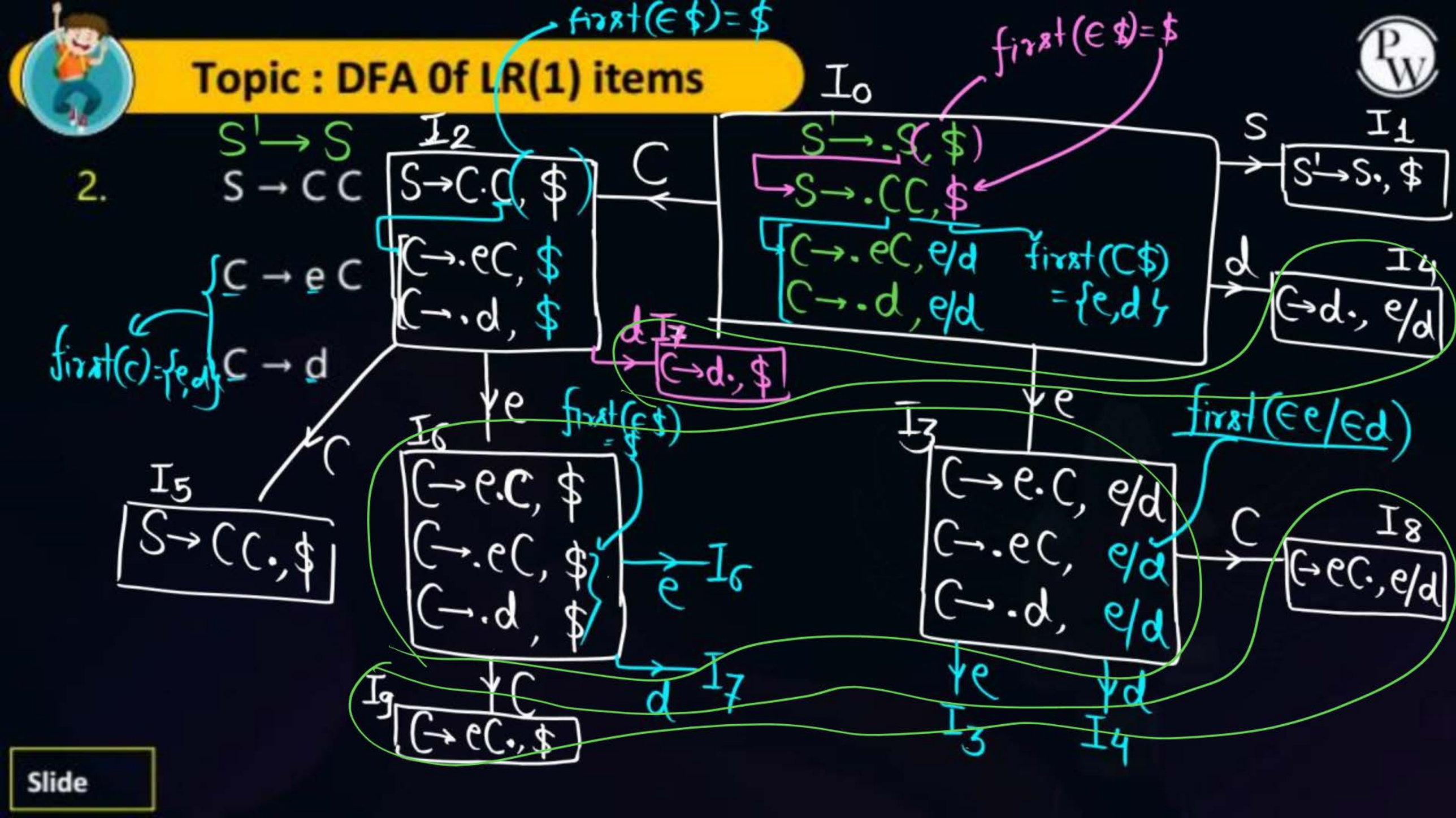
Topic : GOTO Part w.r.t. LALR(1) Parser



- If there exists a transition out of state "i" on non-termina X to state "j" , then

  GOTO [i , X] = j

Topic : DFA Of LR(1) items





Topic : Construction of LALR(1) Parser table

Construct the LALR(1) parsing table for the grammar.

① S→CC

② S→eC

③ C→d

	Action			GOTO	
State	e	d	$	S	C
0	S_{36}	S_{47}		1	2
1			Accept		
2	S_{36}	S_{47}			5
36	S_{36}	S_{47}			89
47	R_3	R_3	R_3		
5			R_1		
89	R_2	R_2	R_2		

Slide



Topic : LALR(1) Grammar

* In LALR(1) parsing table if no entry in the action part contains more than one option, then corresponding gramman is LALR(1) gramman
* If any entry in LALR(1) parsing table contains more than one option then corresponding gramman is not LALR(1) gramman

Slide

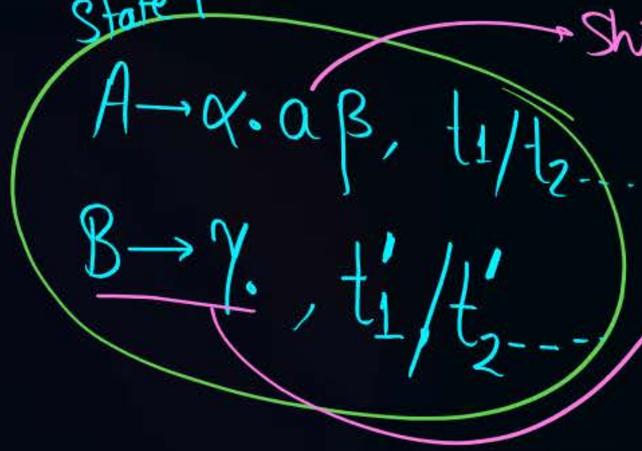
* Check for SR Conflict or RR Conflict Only after merging the States with same Core set.



Topic : SR Conflict w.r.t LALR(1) Parser

{Same as CLR(1), but we check only after merging the states}

SR Conflict exists in State 'i' w.r.t. LALR(1) parser only if State 'i' Contains items of the form



And

$a \in \{t_1', t_2', \ldots\}$

SR Conflict only if $a \in \{t_1', t_2', \ldots\}$

Slide



Topic : RR Conflict w.r.t LALR(1) Parser

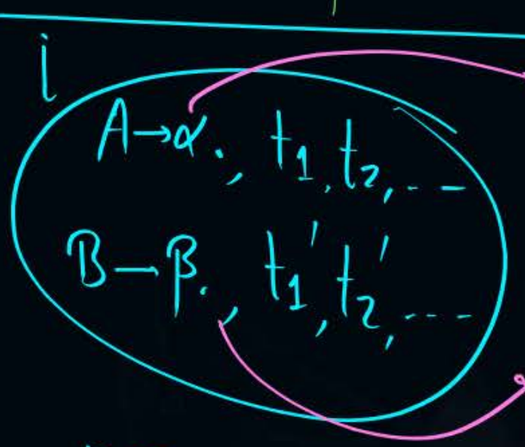
RR Conflict exists in state 'i' w.r.t. LALR(1) parser only if state 'i' contains items of the form,

$A \rightarrow \alpha.$, $t_1/t_2, \ldots$

$B \rightarrow \beta.$, $t_1''/t_2' \ldots$

and

$\{t_1, t_2, \ldots\} \cap \{t_1', t_2', \ldots\} \neq \emptyset$



Reduce at $t_1, t_2, \ldots$

Reduce at $t_1', t_2', \ldots$

RR Conflict exist only if

$\{t_1, t_2, \ldots\} \cap \{t_1', t_2', \ldots\} \neq \emptyset$

Slide

* State with SR or RR Conflict w.r.t. LALR(1) Parser is called inadequate State w.r.t. LALR(1).

→ If any inadequate State exists w.r.t. LALR(1) then corresponding grammar is not LALR(1) grammar, otherwise LALR(1) grammar.



Topic : Parsing algorithm w.r.t LALR(1)

Same for all LR parsers.

Slide



Topic : Note

(1) If Core-set of all states of the dfa of LR(1) items are different, then we say that the dfa of LR(1) items is already minimized. And in that parser table of LALR(1) will be exactly same as parser table of CLR(1)

And also the number of states in dfa of LR(1) items will be exactly same as number of states in dfa of LR(0) items for that gramma

Slide



Topic : Note

(2) If Core-set of two or more states in the dfa of LR(1) item is same, then states with the same Core-set can be merged into a single state.

* And in that case parser table of LALR(1) will be different from the parser table of CLR(1)

And also the number of states in dfa of LR(1) items will be more than the number of states in dfa of LR(0) items for that grammar

Slide

③ Let n_1, n_2, n_3 and n_4 are the number of states in the parser table of LR(0), SLR(1), LALR(1) and CLR(1) respectively,

i.e. Parser → LR(0) SLR(1) LALR(1) CLR(1)

States → n_1 n_2 n_3 n_4

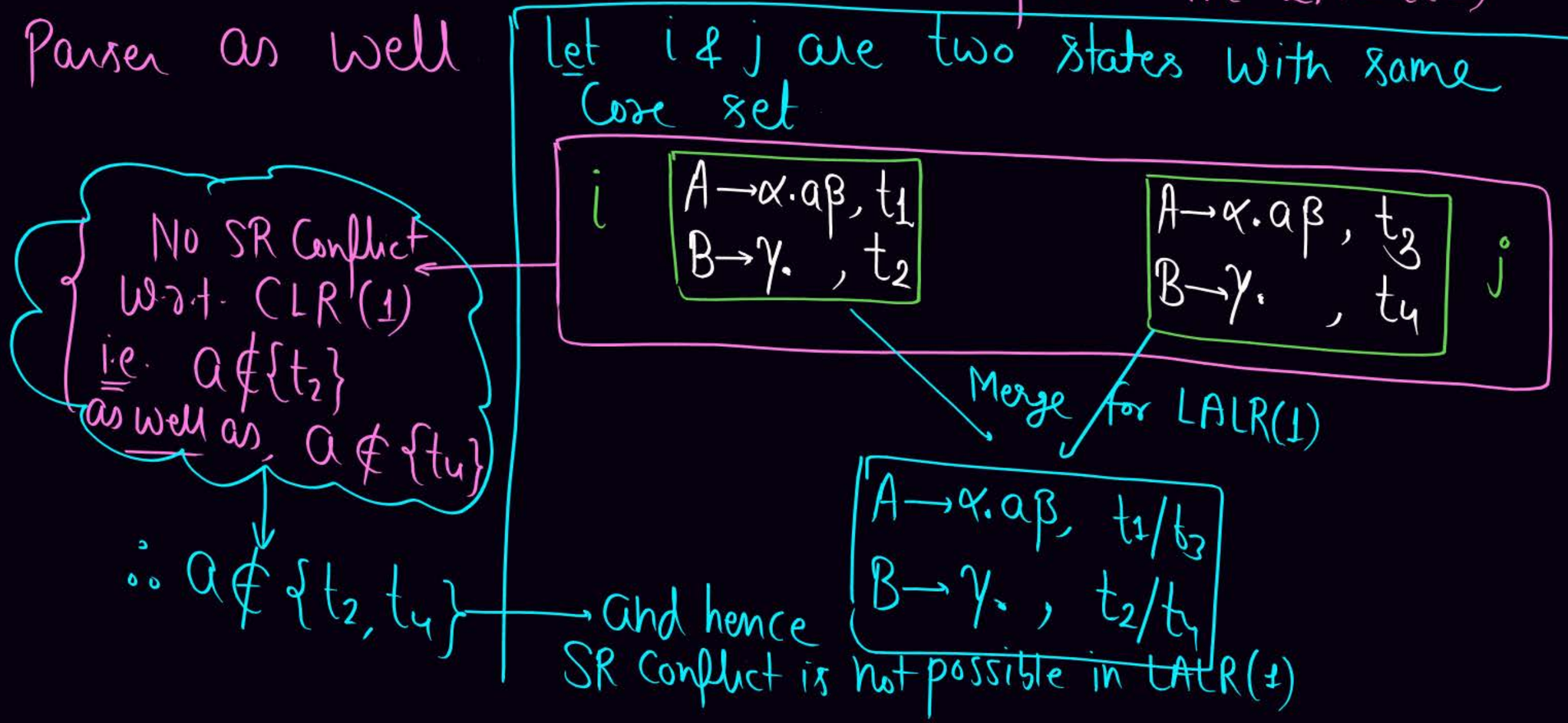
then in general,

$(n_1 = n_2 = n_3) \leq n_4$

- ≤ → when dfa of LR(1) is not already minimized (i.e. there are states with same Core-set)
- = → when dfa is already minimized

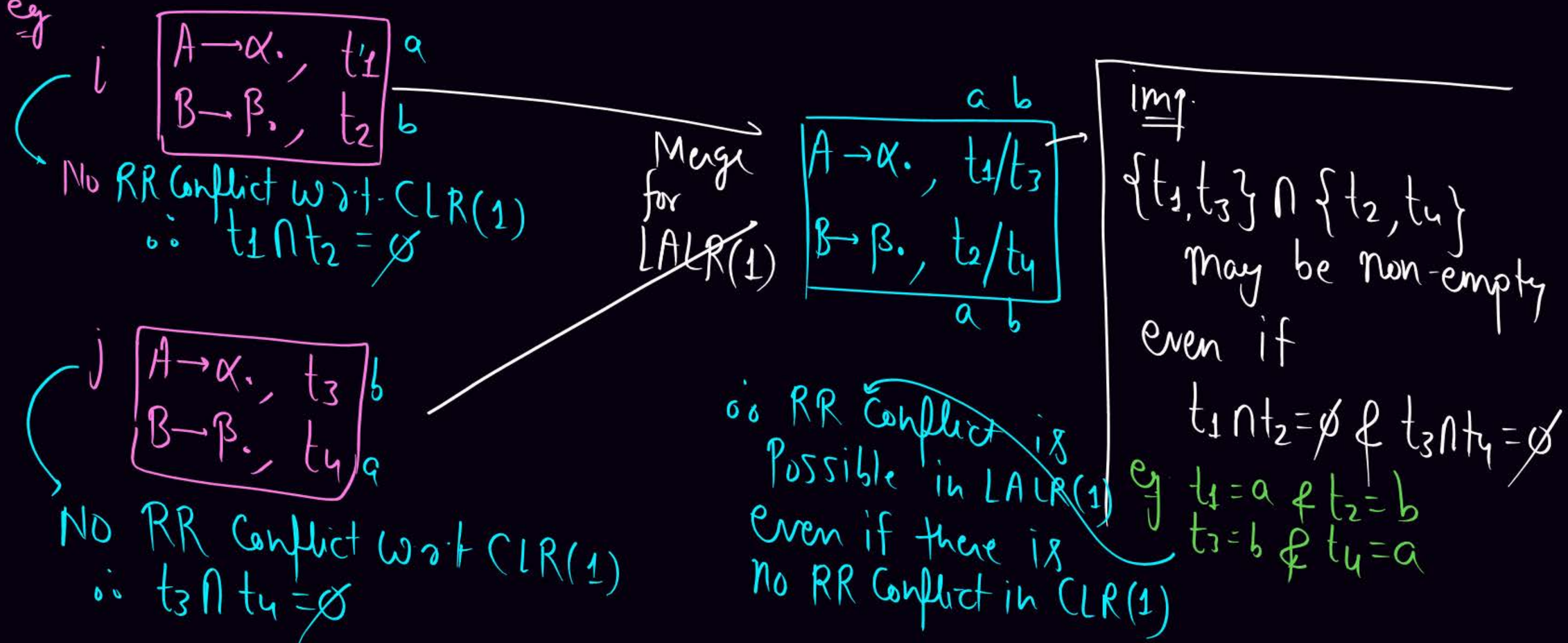
IMP Note:

④ If there is no SR Conflict in CLR(1) parser, then there can not be SR Conflict in LALR(1) Parser as well



Imp Note:

⑤ RR Conflict may be present in LALR(1) parser even when these is no RR Conflict in CLR(1) parser



Note:

⑥ If grammar is not a CLR(1) grammar, then it can never be LALR(1) grammar.

⑦ If grammar is CLR(1) grammar, then it may or may not be LALR(1).



2 mins Summary

Topic LALR(1) Parser

Topic LALR(1) Grammar

Topic Comparison between number of states

Operator Precedence Parser (Tomorrow)

Slide

THANK - YOU